Information Sciences

Vrije Universiteit Amsterdam - Faculty of Science - Information Sciences - 2025-2026

Information Sciences is a multidisciplinary area spanning Information and Communication Technology (ICT) and its practical application in society. Are you interested in how information is created and processed in companies and organisations? Are you interested in technology and its practical applications? Do you believe it is important to consider the role of people, organisations and cultures in the design, modelling, and sharing of information? If so, then our one-year master programme in Information Sciences might be the right choice for you!

The first semester includes courses building the competences for the digital transition and sustainability like Digitalization and Sustainability, Knowledge Organisation, and Software Architecture, as well as a methodological course on Research Methodology and Thesis Design. The second semester focuses on an individual research project, next to which students choose two electives to specialise in competencies for entrepreneurship and IT consultancy, or digital society and consultancy for sustainability.

Throughout the entire programme, the Portfolio Digital Transition and Sustainability guides students to reflect on the lessons learned in the courses and build their own IS portfolio.

Graduates from the Master in Information Sciences typically move on to pursue careers as a.o. IT consultants/Data Analysts, Digital Architects, and Digital Sustainability Officers. They work at leading ICT companies; highly digitalized companies in other sectors, such as Energy, Banking, Transportation and Logistics; and the IT departments of public sector organizations, such as Ministries and Municipalities.

Info

Level	Master
Language	English
Duration	1 year
Type	Full Time
Credits	60 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
More information about this programme

Information Sciences expired courses 4
Information Sciences suggested elective courses 4
Information Sciences compulsory courses 4
Courses 4
Digital Architecture 4
Digital Business Ecosystems and Platforms 6
Digitalization and Sustainability 8
Entrepreneurship in Analytics and AI 9
Ethics and Privacy 10
Fair Digital Design 12
ICT for the Global South 13
Inclusive ICT 14
Knowledge Organization 15
Master Project Information Sciences 16
Research Methodology and Thesis Design 18
The Social Web 19

Information Sciences expired courses

Information Sciences suggested elective courses

Description

You must choose a total of two electives in period 4, period 5 or period 6. If you choose an elective that is not pre-approved, you need to request approval from the Examination Board in advance.

Courses

Course Name	Period	Credits	Code
Digital Business Ecosystems and Platforms	P4	6EC	XM_0176
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Fair Digital Design	P4	6EC	XM_0132
Ethics and Privacy	P5	6EC	XM_0087
ICT for the Global South	P5	6EC	XM_0142
Inclusive ICT	P6	6EC	XM_0177

Information Sciences compulsory courses

Courses

Course Name	Period	Credits	Code
Master Project Information Sciences	Ac. Year (September)	18EC	XM_405083
Digitalization and Sustainability	P1	6EC	XM_0089
Knowledge Organization	P1	6EC	X_405065
Digital Architecture	P2	6EC	XM_0127
The Social Web	P2	6EC	X_405086
Research Methodology and Thesis Design	P3	6EC	X_405085

Courses

Digital Architecture

Course Code	XM_0127
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.C. de Boer
Examiner	dr. R.C. de Boer
Teaching Staff	dr. R.C. de Boer, dr. R.C. de Boer
Teaching method(s)	Seminar, Lecture

Course Objective

Knowledge and understanding
At the end of the course, students will have gained knowledge about architecturally significant requirements, architecture representation, design, quality and analysis. They understand the drivers behind architectural decisions and the ‘wicked’ nature of architecture design, and appreciate the role of the architect.

Applying knowledge and understanding

Students will be able to reason about the architecture of non-trivial software-intensive systems, and apply their knowledge and understanding to a concrete (fictitious) case that involves many stakeholders with different needs and concerns.

Making judgments
Students will make their own architectural decisions, and become aware of the rationale involved in making trade-offs and choosing design options among alternatives. They learn to appreciate the design decisions proposed by others. In a serious game setting, students will further experience architectural design decision making.

Communication skills
Students present their progress to the other course participants, and engage in debate about their architecture design. They communicate their final architecture design, including the architectural decisions and their rationale, in a document.

Learning skills
Students have to coordinate and negotiate team work, and manage a significant workload in the allotted time. Through searching and integrating background material, students build a progressive understanding of the 'problem domain' relevant to the case.

Course Content

Students work in groups to design an architecture for a fictitious software-intensive system. They have to develop different representations (called views) of the architecture. These different representations emphasize different concerns of people or organizations that have a stake in the system. The group work is complemented by theory lectures and in-depth sessions.

Additional Information Teaching Methods

Teaching methods and contact hours:

- Lectures: two hours per week
- Group work with a number of assignments; two contact hours per week
- Tutorials and in-depth sessions: two hours per week.

Attendance of the first lecture is compulsory! Students are required to participate in group work. All teams must be present at their scheduled contact hours to present their progress and engage in debate with the other teams.

Note that this course is fully taught on campus. Remote participation is not possible.

Method of Assessment

Graded assignments

Project assignments (group assignment): There are two assignments based on a practical case. These assignments are done as group work. The average grade of the assignments should be ≥ 5.5 and will account for 75% of the final grade. Your work will be evaluated by means of a rubric published on Canvas.

Written exam (individual assignment): To complete the course you will have to take part in the written exam for Digital Architecture. The exam takes place in the exam week at the end of the period. Your grade for the exam should be ≥ 5.5 and will account for 25% of your final grade.

Pass/fail-assigments

Participation in the serious game and Game survey (individual assignment): In-person participation in the game is compulsory. The game will be played by all students simultaneously on campus. After playing the architecture design decision-making game, each student should fill out the game survey individually. There will be only one opportunity to play the game, it is not possible to retake this assignment on another occasion. The assignment will be graded as "pass" or "fail".

Practical sessions: attendance (group assignment) and presentation (individual assignment): Participation in the practical sessions is compulsory. Every team must be present each week and present their progress and engage with other teams, and all team members are expected to be there. Every team member must present the team's progress at least once. The assignment will be graded as "pass" or "fail".

Weekly progress report (individual assignment): We ask you to – individually – complete a simple weekly survey to indicate your team's progress, collaboration and communication and the participation of all team members (including yourself). The assignment will be graded as "pass" or "fail".

Compensation and final grade

Both the exam grade and the weighted average of the project assignment grades MUST be 5.5 or higher. Compensation between the two is not possible. Additionally, all "pass"/"fail"-assignments and presentations must be "pass" in order to pass the course.

- In case any of the pass/fail-assignments is a 'fail', the final grade of the course will be 'NVD', otherwise:
- In case any of the graded assignments is insufficient, the final grade of the course will be that insufficient grade, otherwise:
- In case all pass/fail-assignments are 'pass' and all graded assignments are at least 5.5, the final grade of the course will be the weighted average of the project assignment grade (75%) and the exam grade (25%).

Mode of re-examination

A failed project assignment can be revised and resubmitted once; a failed presentation can be retried once; failing or missing the game assignment CANNOT be retried. A failed exam can be retried in the resit period.

Literature

Len Bass et al., Software Architecture in Practice, 4th edition, 2021

Additional Information Target Audience

Master Information Sciences
Master Computer Science

Custom Course Registration

Group enrolment takes place in Canvas.

Explanation Canvas

Further information for this course will be made available online (Canvas). All students must be enrolled in the course Canvas community.

Recommended background knowledge

Software design.

Digital Business Ecosystems and Platforms

Course Code	XM_0176
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J. Gordijn
Examiner	dr. J. Gordijn
Teaching Staff	dr. J. Gordijn
Teaching method(s)	Seminar, Lecture

Course Objective

An ecosystem is a collection of parties who work cooperatively and competitively to satisfy customer needs. Well-known examples are Google, Facebook, and Netflix, but there are many other eco-systems in the field, e.g. in Fintech, energy, and automotive industry. In this course, we focus on digital ecosystems which are enabled by Internet- and related technologies. Frequently, these ecosystems are disrupted by innovative information technologies, such as machine learning, blockchain, and 5G mobile. The overall goal of this course is to analyze disruptive opportunities for an eco-system given a particular emerging technology. This year, that technology is blockchain.

Upon completion of this course, students will:

- have gained knowledge and understanding of the concept of ‘eco-system’ (Knowledge and understanding);
- have gained knowledge and understanding of various methods for designing and analyzing eco-systems and their e-services, including e3value, trust modelling, BPMN, and UML class modelling (Knowledge and understanding);
- have gained knowledge and understanding of the selected technology of this year in order to analyze, design and implement a disruptive eco-system (Knowledge and understanding);
- be able to analyze and design an eco-system offering an e-service from different viewpoints (Applying knowledge and understanding);
- be able to elicit different e-service (business) ideas that can be provided in an eco-system (Applying knowledge and understanding);
- be able to explore the e-service idea from a business perspective. In this way, the student will be able to further design and develop the e-service idea using different analysis techniques (Applying knowledge and understanding);
- be able to transform the business perspective of the e-service design into a design reflecting the IT perspective (Applying knowledge and understanding);
- be able to implement (part of) the e-service using the selected disruptive technology (Applying knowledge and understanding);
- be able to make informed decisions about eco-systems and their e-services based on possible risks and opportunities (Making judgments);
- be able to assess the quality and correctness of the various conceptual models created (Making judgments);
- be able to pitch a new idea for an eco-system/e-service to investors (Communication skills);
- be able to concisely present the selected solution (Communication skills);
- be able to write a scientifically well-founded report about the eco-system/e-service under development in a group of students (Communication skills);
- have learned to work in a project group and keep each other accountable for agreed results (Learning skills);
- have learned to improve themselves based on various feedback mechanisms (Learning skills).

Course Content

The course comprises the following topics:

- Eco-systems and platforms
- Introduction in the selected technologyof this year
- Business models and value proposition design
- The e3value method for understanding business networks
- The Business Process Modelling Notation for analysing cross-organizational networks
- The Unified Modelling Language – class modelling
- Transaction modelling
- Technological design issues

Additional Information Teaching Methods

The following methods are used:

- Lectures twice a week. These lectures include in-class workshops to practice the various methods used in this course
- Group work: teams of about 2/3 students design an eco-system. At the beginning of the course a number of eco- systems are offered as inspiration
- Group feedback sessions: Each week, students will get personal feedback in face-to-face feedback sessions. Moreover, groups will receive feedback on their submitted reports by email.

Method of Assessment

Individual exam on the theory (50%), Teamwork project assignments (50%).

To pass the course, the scores of each component must be 5.5. or higher.
Compensation: it is not possible to compensate one component with another.
Mode of re-examination: if insufficient, each component can be revised and resubmitted once.

Literature

A (selection of a) list of papers on the course topics. Material distributed on Canvas.

Additional Information Target Audience

Master Artificial Intelligence
Master Computer Science
Master Information Sciences

Custom Course Registration

Further information for this course will be made available online (Canvas).
All students must be enrolled in the course Canvas community. Group enrolment takes place in Canvas.

Recommended background knowledge

The student should have sufficient knowledge about Requirements Engineering and Conceptual Modelling and should master such a modelling method (at least UML class or activity diagrams).

Digitalization and Sustainability

Course Code	XM_0089
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. P. Lago
Examiner	prof. dr. P. Lago
Teaching Staff	prof. dr. P. Lago
Teaching method(s)	Seminar, Lecture

Course Objective

At the end of the course, the students will:

- be familiar with basic knowledge about digitalization and digital transformation, sustainability, and the role of digitalization in achieving business- and other sustainability goals within society and organizations (Knowledge and understanding; Applying knowledge and understanding).
- be able to reason about the technology- and business-related digitalization and sustainability concerns, and apply their reasoning to a concrete project (Applying knowledge and understanding; Making judgments).
- have a basic understanding of the types of sustainability impacts of digital solutions, so to identify and assess the trade-offs between the different sustainability concerns addressed by digital solutions (Knowledge and understanding; Applying knowledge and understanding; Making judgments).
- be able to write a scientific report about a concrete digitalization-and-sustainability project in a group of students (Applying knowledge and understanding; Making judgments; Communication skills).
- be trained to (i) explore the problem- and solution space in the digital transition of a specific sector/domain, and (ii) identify and address a set of relevant sustainability goals (Applying knowledge and understanding; Making judgments; Learning skills).
- be able to reflect critically on what learned, its impact in society, and express own motivated opinions (Applying knowledge and understanding; Making judgments; Communication skills; Learning skills)

Course Content

The course follows a mixed-method approach and includes live lectures, flipped-classes with video lectures, active discussions, and teamwork.

The lectures explain the basic concepts related to digitalization (such as the notions of digitalization and digital transformation, the role of technology, and the impact on business and society) and sustainability (such as the notions of sustainable software and software for sustainability, how to frame sustainability-quality concerns in the design of digital solutions, and how to assess sustainability impacts).

Discussions are based on the video/live lectures and a set of papers the students will reflect upon.

The students participate in small teams to incrementally develop an understanding of the digital transformation of a selected sector/domain and the related sustainability concerns, and work on a shared project and associated report.

Additional Information Teaching Methods

The course will be centered on the practical project (in teamwork) of analyzing the sustainability-, business- and technology- implications of digitalizing a sector or application domain of own choice. The project is generic by design, to allow the students reflect on the gaps and opportunities for digital transformation, and choose their own

focus for the most promising digitalization and sustainability goals.

The theory will be explained in lectures (3 on digitalization topics and 3 on sustainability topics) some of which may be video-recorded upfront. Each week, live lectures will engage and challenge the students in lively discussions; while for flipped-classes the students will watch the video lecture upfront and individually carry out simple understanding tasks which will be discussed afterward in a live plenary session. The students will also read a paper each, distributed upfront related to each week's topics, and answer reflective questions that help apply the gained knowledge to the practical project. There will be weekly active discussions in teamwork sessions, where the students will work in small teams on their practical project. Input to the teamwork sessions will be the reflection on the papers the team has read, and the specific project assignment. There is also a significant amount of self-study, both to study the theory, read the papers, and work on the practical project. The scheduled sessions are 2 hours per week lectures/discussion (l), and 2 hours per week teamwork (s).

Attendance is mandatory.

Method of Assessment

Type and Weight of each component: Teamwork project assignments (80%, two group assignments), Individual assignment (20%). As part of the teamwork project assignments, each student gives an individual presentation. To pass the course, the scores of each component must be 6.0 or higher.

Compensation: it is not possible to compensate one component with another.

Mode of re-examination: if insufficient, each component can be revised and resubmitted once.

Literature

A (selection of a) list of papers on the course topics. Course material distributed on Canvas.

Additional Information Target Audience

MSc Information Sciences, MSc Computer Science

Additional Information

The course has a maximum enrolment limit of 100. The students of the Master Information Sciences, for whom it is compulsory, have priority over the students that take this course as an elective or a constrained choice course.

The students that take this course as an elective or a constrained choice course run the risk of not being able to participate, even if they have registered on time, if the maximum enrolment limit has been reached.

The students of the Master Information Sciences, for whom it is compulsory, run the risk of being initially refused if the enrolment limit has been reached at the time of registration. However this will be adjusted manually after registration has been closed so that all the students of the Master Information Sciences will be able to participate.

Custom Course Registration

Further information for this course will be made available online (Canvas). All students must be enrolled in the course Canvas community.

Group enrolment takes place in Canvas.

Explanation Canvas

Instructions about group enrolment and (if applicable) software tools for attendance will be provided at least one week before the course starts. All students must be enrolled before then.

Recommended background knowledge

Basic knowledge of some programming language. Knowledge of UML.

Entrepreneurship in Analytics and AI

Course Code	XM_0090
Credits	6
Period	P4
Course Level	400

Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. Stam
Examiner	prof. dr. W. Stam
Teaching Staff	prof. dr. W. Stam, H.V. Nguyen, drs. G.J. Beekman MA, P. Afonso Branco Ramos Pinto
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

Academic and research skills: Students are able to generate a promising new business idea in the field of AI & Analytics and develop a prototype that validates the proposed customer needs and market potential.

Bridging theory and practice: Students understand the core theories and concepts in the field of entrepreneurship, as reflected by their ability to analyze, compare, and apply these theories and concepts to analyze and solve the key management challenges faced by real-world entrepreneurs.

Entrepreneurial soft skills: Students are able to effectively work in a team for generating and developing a new venture idea and can pitch the business concept to external stakeholders in a professional, engaging manner.

Course Content

Entrepreneurship in Analytics and AI is an engaging course for students who like to learn about entrepreneurship, its role and importance in our society, and the process by which entrepreneurs transform new ideas into successful AI-driven business ventures.

In this course, students will learn the ropes of this emerging field via three educational pillars:

1. Students will learn to understand and apply theories from economics, sociology, and psychology to study key topics in entrepreneurship, including innovation, new venture creation, entrepreneurial competences, entrepreneurial finance, scaling and growth of new ventures, corporate entrepreneurship, and entrepreneurial ecosystems. Theoretical understanding of these subjects is applied to real world cases focusing on key management challenges faced by entrepreneurs and how to creatively overcome these challenges.
2. Students will familiarize with a set of entrepreneurial soft skills, which they will put into practice when they will be asked to approach different stakeholders to further their team projects to gain commitments or to obtain important information. For this purpose, the students will receive training in generating initial ideas for opportunities as well as networking and network utilization.
3. Students will be introduced to the actions and strategies that promote the process of new venture creation. For this purpose, the students, supported by the lecturers and coaching, will write (and pitch) a Student Design Sprint report (including prototype) around an innovative new venture opportunity in the field of Analytics and/or AI that they deem interesting for entrance into the market.

Additional Information Teaching Methods

During this course, students will attend both (guest) lectures and tutorials. The lectures explain relevant concepts, theories, and frameworks in the field of entrepreneurship and illustrate their usefulness using various practical examples. The tutorials are designed to facilitate students in making progress with their team projects by providing appropriate guidance and feedback.

Method of Assessment

The final grade for this course is based on a written individual exam, several individual pass/fail assignments, and a Student Design Sprint (SDS) project assignment.

The written exam and pass/fail assignments are individual assessments whereas the SDS assignment is a team assessment. To pass the course, the exam and SDS assignment should be graded as sufficient. Both forms of assessment have one option for re-sit in case students received an insufficient score.

Literature

The literature for this course constitutes a selection of academic articles and case studies.

Ethics and Privacy

Course Code	XM_0087

Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A.E. Guzman Ortega
Examiner	dr. A.E. Guzman Ortega
Teaching Staff	dr. A.E. Guzman Ortega
Teaching method(s)	Study Group, Lecture

Course Objective

At the end of the course students should be able to:

- Identify ethical and social issues in information systems (Knowledge and understanding)
- Reason about ethical and social issues in information systems (Applying knowledge and understanding) (making judgements)
- Design ethically and socially responsible information systems (Applying knowledge and understanding)
- Communicate a reasoned ethical position (Communication skills)

Course Content

The lecture will cover an introduction to ethical thinking and ethical issues faced by current Information Technology. Specifically, we will cover the interplay of ethics and information systems, social issues caused by poorly handled ethical concerns, and privacy in information technology. We will emphasize areas in which Information Technology has raised societal and ethical challenges in recent years: privacy, algorithm bias, online content dissemination, and accountability in the design of Information Technology.

During the course, students will reflect through actual real-world examples on the main ethical issues faced by Information Technology and will design and present potential alternatives for identified ethical concerns.

Additional Information Teaching Methods

The lecture will mostly be in a seminar format. Students will read and watch material before the lecture, write short reports on the material and discuss the main points during the lecture with the rest of their peers and lecturers. Students will also present (in groups) a topic of their choice within the course domain and guide the discussions among the rest of their peers. During these presentations each student will be asked questions about the material they present. The quality of their answers will be part of the final grade.

Method of Assessment

Assignment #1 (group): 30%
Assignment #2 (group): 30%
Assignment #3 (individual): 30%
Participation in class: 10%
6 pass/fail homeworks

To pass the course, 5/6 pass/fail homework assignment need to be passed.
Mode of re-examination: Assignment #1 and #3 can resubmitted once (only if submitted and failed at the first try). The topic of this resubmission will be announced by the end of the lecture and will not be the same one of the first submission.

Assignment #2 (consisting of a presentation) cannot be compensated. Each student will individually be asked questions about the material covered during the presentation. The quality of these answers will contribute to the grade of the individual student.

Literature

Relevant readings will be made available in Canvas.

Additional Information Target Audience

MSc Information Sciences

Recommended background knowledge

Basic programming experience. Familiarity with Information Systems.

Fair Digital Design

Course Code	XM_0132
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.M.A.L. Beauxis Aussalet
Examiner	dr. E.M.A.L. Beauxis Aussalet
Teaching Staff	prof. dr. P. Lago, dr. E.M.A.L. Beauxis Aussalet
Teaching method(s)	Seminar, Lecture

Course Objective

At the end of the course, students should be able to:

- Identify fairness issues in digital solutions (Knowledge and understanding)
- Reason about fairness issues in digital solutions (Applying knowledge and understanding)
- Design fair digital solutions (Applying knowledge and understanding)
- Communicate a reasoned position with respect to fairness (Communication skills)
- Reflect critically on what was learned; discuss and express motivated opinions (Learning skills)

Course Content

In this course, students study scientific research from complementary fields within Computer Science, Information Sciences, and related disciplines. The course entails a mix of invited research lectures and invited talks from practicing experts. Both elaborate on topics around fair digital design. Such topics will be further researched by the students through collaborative learning activities.

Working individually and in groups, students analyse the social, technical, environmental, and economical components of fairness issues in digital solutions. Students model fairness issues as a software quality concerns, using the Sustainability Assessment Framework (SAF) Toolkit.

Working in groups, students read and discuss research articles exploring a selected topic. The discussion involves the content of the publications but also the scientific contributions, quality of the work, and relevance to society. Finally, student groups submit a 4-page position paper in which they elaborate on their position with respect to the selected topic, building upon the read literature and the reflections on the course material.

Additional Information Teaching Methods

Lectures (with interactive elements). Seminars (including presentation and discussion sessions). In-person attendance is mandatory.

Method of Assessment

Type and Weight of each component: Teamwork assignments (70%, position paper assignment; 30% digital design assignment), both with some individual components. To pass the course, the scores of each assignment must be 6.0 or higher. In case of unclear individual contribution on any of the assignments, the teaching staff may ask for a clarification with an oral check. As part of the teamwork assignments, each student gives an individual presentation. Active participation in class is also expected, with a pass/no-pass assessment.

Compensation: it is not possible to compensate one component with another.

Mode of re-examination: if insufficient, each component can be revised and resubmitted once.

Literature

Collection of articles.

Additional Information Target Audience

MSc Information Sciences

Custom Course Registration

Further information for this course will be made available online (Canvas). All students must be enrolled in the course Canvas community. Group enrolment takes place in Canvas.

Explanation Canvas

Instructions about group enrolment and (if applicable) software tools for attendance will be provided at least one week before the course starts. All students must be enrolled before then.

Recommended background knowledge

Course Digitalization and Sustainability.

ICT for the Global South

Course Code	XM_0142
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. Aslam
Examiner	dr. K. Aslam
Teaching Staff	dr. J. Gordijn, dr. K. Aslam
Teaching method(s)	Seminar, Lecture

Course Objective

At the end of the course, students will be able to:

- Gain a comprehensive understanding of the digital divide, challenges and opportunities in the Global South and the role of ICT in addressing them. (Knowledge and Understanding)
- Apply technical skills and innovative approaches to develop innovative ICT solutions to meet specific needs in the Global South. (Applying Knowledge and Understanding)
- Evaluate the impact of ICT policy and governance on socio-economic development and how it contributes(d) to digital divide by considering ethical, cultural, and sustainability factors. (Making Judgements)
- Effectively communicate project design and outcomes both orally and in academic reports. (Communication)
- Develop the capacity for self-directed learning about emerging technologies and their potential applications in ICT for the Global South. (Lifelong Learning Skills)

Course Content

This course explores the role of Information and Communication Technologies (ICTs) in addressing socio-economic challenges faced by developing countries, commonly referred to as the Global South. It provides a comprehensive understanding of how ICTs can be harnessed to foster development, improve governance, enhance education, healthcare, and empower communities.

The lectures (in the form of interactive sessions) will explore how technological innovations can meet specific local needs through sustainable and culturally relevant solutions. Students will engage in hands-on projects to apply their technical skills and innovative thinking to develop real-world ICT solutions tailored to the nuanced demands of the Global South.

Students will be guided during the course to develop a reflective practice that includes empathy and ethical considerations, essential for developing ICT solutions for low resource environments. The course focuses that ICTs are powerful enablers for achieving many of the United Nations Sustainable Development Goals (SDGs), teaching students how to leverage technology to address some of the world’s most pressing challenges.

Additional Information Teaching Methods

This lectures will be delivered in an interactive seminar-style format. Prior to each class, students are expected to engage with assigned readings and multimedia resources to build a foundational knowledge base. During the lecture, students will discuss main insights and reflect on their own viewpoint in relevance to the pre-class materials.

Method of Assessment

During the course students will work in a group of 3-5 (depending on the total number of students) to derive a solution for a particular problem in the context of Global South. The projects will be research-based and along the way students will need to submit four assignments in total:

Assignment #1: group (a written report of overall description of project stating goal and research questions) - 15%
Assignment#2: group (a written report of detailed description of methodology and plan to conduct the study) - 15%
Assignment#3: group - (a written report of results achieved, discussion of results in the specific context for which study is conducted and future outlook of the project) 40%
Assignment#4: individual oral presentations - 20%
Attendance: 10%

Each of the assignments should be at least graded with a 5.5 or higher and the total marks should be 6 or higher for a student to pass the course. One resubmission is allowed for Assignment#1 and Assignment#2 in case of achieved grade below 5.5. There is no resubmission for Assignment#3 and Assignment#4.

There will be no final exam.

Literature

Relevant literature will be made available on Canvas.

Additional Information Target Audience

MSc AI, MSc CS, MSc IS

Recommended background knowledge

We expect students to have at least a basic knowledge of programming and how ICT projects are conducted. Familiarity with mixed-methods research or design science will be a plus, but not mandatory.

Inclusive ICT

Course Code	XM_0177
Credits	6
Period	P6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. Aslam
Examiner	dr. K. Aslam
Teaching Staff	dr. K. Aslam, dr. J. Gordijn
Teaching method(s)	Study Group, Lecture

Course Objective

At the end of the course, students will be able to:

1. Develop an understanding of the socio-economic characteristics of marginalized and minority populations in the global North and the disparities in ICT access and usage. (Knowledge and Understanding)
2. Elicit the specific requirements and limitations of marginalized populations and then propose tailored ICT solutions for them. (Applying Knowledge and Understanding)
3. Analyze the potential impact and ethical implications of the designed ICT solutions on the target populations. (Making Judgements)
4. Effectively communicate the outcomes of ICT needs assessments and solution proposals both in written reports and oral presentations. (Communication)
5. Engage in continuous learning about the evolving needs of marginalized populations in relation to ICT applications and maintaining an adaptive approach to problem-solving in these cases. (Lifelong Learning Skills)

Course Content

This course will provide students with an understanding of the global North's socio-economic landscape and the marginalized and minority populations within it (e.g., differently-abled people, migrants and refugees, elderly

population, ethnic minorities etc.) exploring disparities in ICT access. The course objectives closely align with several of the United Nations Sustainable Development Goals (SDGs) such as Reduced Inequalities (SDG 10), Sustainable Cities and Communities (SDG 11), and Peace, Justice, and Strong Institutions (SDG 16).

We will discuss methods for assessing ICT needs for these populations through case studies and community engagement activities. Students will learn about designing inclusive ICT solutions also by working together in groups where they will propose and present a tailored ICT application. The course will also incorporate discussions on the effectiveness of public-private partnerships and policy implementations in fostering ICT accessibility in the Global North.

Additional Information Teaching Methods

The lectures will be delivered in an interactive seminar-style format. Prior to each class, students are expected to engage with assigned readings and multimedia resources to build a foundational knowledge base. During the lecture, students will also reflect on their own viewpoint in relevance to the material discuss in class.

Method of Assessment

Two group assignments: 60%
Four pass/fail homeworks (individual, may also include delivering recorded presentation): 30%
Attendance: 10%

Each of the assignments should be at least graded with a 5.5 or higher and the total marks should be 6 or higher for a student to pass the course.

Literature

Literature will be made available on Canvas.

Additional Information Target Audience

MSc CS, MSc AI, MSc IS

Knowledge Organization

Course Code	X_405065
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.M. Siebes
Examiner	dr. R.M. Siebes
Teaching Staff	dr. V. de Boer, dr. R.M. Siebes, T. Kuhn
Teaching method(s)	Seminar

Course Objective

At the end of this course, students will:

- have gained high-level knowledge of the potentials and problems with formal knowledge organization (knowledge and understanding);
- have gained high-level knowledge of the possible solutions and applications for formal knowledge organization (knowledge and understanding);
- be able to understand concrete knowledge organization methods and practically apply them (knowledge and understanding, apply knowledge and understanding);
- be able to communicate own thoughts and reflections on the topic of knowledge organization (communication)

Course Content

This course covers the general principles and methods that form the foundation of information organization and knowledge-intensive processes, as well as the contexts in which they can be applied and the interaction with users. The lecture topics include knowledge modeling, ontologies, logic, controlled natural language, Semantic Web and Linked Data, as well as knowledge maintenance and evaluation, in addition to guest lectures on specific

applications and domains.

Additional Information Teaching Methods

Lectures are in a hybrid setup and can be followed both in the lecture room and online, and will be video-recorded. These recordings are made available through Canvas afterwards.

Method of Assessment

Final written exam (75%), final assignment (25%). All parts are individual (no group work). To pass the course students need a passing grade (5.5 or more) both for the exam and for the final assignment. There will be a resit for the written exam and for the final assignment. For the final assignment, the maximum resit grade is 6.0.

Literature

Papers to read will be announced through Canvas.

Additional Information Target Audience

MSc Artificial Intelligence
MSc Computer Science
MSc Information Sciences

Additional Information

The course has a maximum enrollment limit of 120. The students of the Master Information Sciences, for whom it is compulsory, have priority over the students that take this course as an elective or a constrained choice course.

The students that take this course as an elective or a constrained choice course run the risk of not being able to participate, even if they have registered on time, if the maximum enrollment limit has been reached.

The students of the Master Information Sciences, for whom it is compulsory, run the risk of being initially refused if the enrollment limit has been reached at the time of registration. However this will be adjusted manually after registration has been closed so that all the students of the Master Information Sciences will be able to participate.

Master Project Information Sciences

Course Code	XM_405083
Credits	18
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	Z. Inayat
Examiner	Z. Inayat
Teaching Staff	dr. I.G. Gerostathopoulos, Z. Inayat
Teaching method(s)	Lecture

Course Objective

The Master Project is the culmination of the Information Sciences Master's programme.

During the project, students will:

1. Develop a deep understanding of the specific aspects that are at the core of the thesis topic (Knowledge & understanding)
2. Develop scientific and professional skills (Lifelong learning skills)
3. Demonstrate the ability to incorporate an element of originality or creativity into their research, for example, in performing a design task or in contributing to the solution or the analysis of a scientific problem; (Apply knowledge and understanding) (Make judgements)
4. Be able to interact with other academics or professionals from industry (and possibly with other students); (Lifelong learning skills) (Communication)
5. Develop the ability to plan a project (Apply knowledge and understanding) (Make judgements)
6. Develop the ability to document and present the final results. (Apply knowledge and understanding) (Communication)

7. Judge and engage in critical discussion about the digital transformation and sustainability in society and business dimensions, threats, and opportunities (Applying knowledge and understanding)(Making judgments).
8. Reflect on the knowledge and competence acquired during the Master's program (Learning skills).

Course Content

The Master Project IS concludes the Master IS programme. In the Information Science programme, the master's thesis course has the following two main parts:

A. Master Thesis:

The master's thesis can be carried out:

1. As an internal graduation project under the supervision of one of the research groups of the Department of Computer Science. A master's thesis can be done as a DUO thesis or individually. in case of DUO thesis, the individual contribution must be clearly identified.
2. As an internship thesis within a professional organisation. A formal agreement (UNL) is signed by all parties between VU and the respective organisation to conduct the internship thesis.

For additional information and rules, we refer to the website of the Faculty of Science. There, you will also be able to find links to the web pages of the research groups of the Department of Computer Science, with options for master's projects.

B. (DTS) portfolio:

This course also includes the Digital Transition and Sustainability (DTS) portfolio that the students keep throughout the entire Master programme. In this portfolio they reflect on the education followed in relation to the principles and practice of the Master Information Sciences. To create the portfolio, the students are given a set of guidelines and reflective questions. These are used each teaching period to reflect on the courses they have just attended and to identify and analyze challenges and solutions related to the topics of digital transition and digital sustainability. The portfolio and the peer groups are in line with constructivist principles whereby students themselves—but together—construct the knowledge and coherence around their package.

Additional Information Teaching Methods

Master Thesis:

For the master's project, the students need to find a project with a VU staff member as their supervisor by the end of period 2. The department shares a list of available projects and supervisors via Canvas each year before the start of the Master thesis. The students select a supervisor and thesis topic either from the shared list or they may come up with their own idea. The project idea is discussed and agreed upon between students and supervisor before the formal start of the thesis. Students will then work on their projects individually and interact with their supervisor in regular meetings (individually or in groups), as agreed with the supervisor. There are no lectures for the Master IS course, but joint meetings may happen when necessary and are announced via Canvas.

DTS Portfolio:

For the portfolio, the students will follow an introductory lecture in the first teaching period. The portfolio will be incrementally built by the individual work of each student which is broken down into six assignments, one for each teaching period. At the end of each teaching period (six times in total), students form peer groups to meet (on-site or online) and discuss progress. The last meeting will be a poster event where students have the chance to present their portfolio to their peers and staff members. The progress of the portfolio will be monitored by a mentor.

Method of Assessment

The final grade of this course corresponds to the grade of the Master thesis. It will be based on the quality of the performed research, the written thesis, the oral presentation, and the student's attitude. These are the weights of the different components:

- Attitude (1/6)
- Research (2/6)
- Thesis (2/6)
- Presentation (1/6)

The final portfolio is submitted together with the thesis. It is evaluated with a pass/no-pass based on the poster presentation (20%) and the assessment of the submitted portfolio (80%). If the student receives a no-pass for the portfolio, she or he can resubmit the portfolio within a 2-month period for a final assessment. The minimum passing grade for each component is 5.5.

Additional Information Target Audience

Master Information Science Students.

Additional Information

You will find useful documentation on all aspects of internships and the Master Project at the website of the Internship Office. This office can also give you advise about internships. It also makes sense to check out the web profiles of potential supervisors, e.g. to gather ideas on projects.

Explanation Canvas

It is advised to join the Master IS Canvas page because the announcements about Portfolio sessions, deadlines, thesis opportunities, and upcoming events for thesis students are made on Canvas. All the deadlines are mentioned for corresponding activities on Canvas.

Research Methodology and Thesis Design

Course Code	X_405085
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Lusini
Examiner	dr. S. Lusini
Teaching Staff	dr. S. Lusini
Teaching method(s)	Lecture

Course Objective

The main goal of the Thesis Design course is for students to acquire basic research methodology skills and apply them to write a research proposal on the subject of their IS Master project.
Upon completion of the course, students will:

- demonstrate the ability to formulate a research design that includes the appropriate research questions and explains how they are answered through the applicable research methods; (Apply knowledge and understanding) (Make judgements)
- be able to argue for their research design with a solid argumentation, explaining the underlying assumptions and pros and cons of the chosen methods; (Communication) (Make Judgements)
- have gained a thorough understanding of the different ways to collect and analyze research data according to the different IS research methodologies; (Knowledge and understanding) (Apply knowledge and understanding)
- be able to identify, describe, and communicate the relationship between the subject of their research and the digital transformation and sustainability themes of the Master IS. (Apply knowledge and understanding) (Make judgements) (Communication) (Learning skills)

Course Content

This course teaches you how to design and carry out scientific research in the domain of information sciences. You will be guided through the different stages of carrying out a research project, from conducting a literature review, developing a research design, to collecting and analyzing data, and writing the findings and discussion section. You will apply this to define your own Master's project and write your own research proposal, which you will carry out during your Master Project Information Sciences.

Additional Information Teaching Methods

Lectures. Individual appointments with Master Project supervisor.

N.B. The course is in period 3, but students are required to sign up for a Master Project/supervisor by the end of period 2. A list of available projects/supervisors will be posted on Canvas in October. Watch out for the announcements on Canvas and make sure you register on time to avoid unnecessary delays!

Method of Assessment

The final grade is based on the grades of the Weekly homework assignments (15%), Oral Exam (25%), Final Research Proposal (35%), and Final Presentation (25%).

All assignments are individual. Students will develop their unique and novel research proposal under the supervision of a staff member. The weekly homework assignments and the Oral Exam are graded by the course teacher. The Final Research proposal and Presentation are graded by the Master Project Supervisor.

A minimum grade of 5.5 for each component is required.

Compensation: It is not possible to compensate one component with another.

Mode of re-examination: If insufficient, each component may be revised and resubmitted once, except for the weekly homework assignments.

Literature

All readings are available on Canvas.

Additional Information Target Audience

MSc Information Sciences

Explanation Canvas

All course materials, readings, and assessment information are posted on Canvas.

N.B. The course is in period 3 but students need to register for a project/supervisor on Canvas by the end of period 2.

Recommended background knowledge

Basic knowledge of qualitative and quantitative research methods.

The Social Web

Course Code	X_405086
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Ceolin
Examiner	dr. D. Ceolin
Teaching Staff	dr. F. Ilievski, dr. D. Ceolin, Z. Inayat
Teaching method(s)	Computer lab, Lecture

Course Objective

In this course, students will:

- Learn theory and methods concerning communication and interaction in a Web context (Knowledge and Understanding).
- Understand how user data is represented and distributed across devices in the context of the Social Web (Knowledge and Understanding).
- Collect and prepare Web data, apply the above skills in practical assignments (Apply Knowledge and Understanding)
- Reflect on social and ethical responsibilities linked to the application of these methods (Making Judgements).
- Communicate knowledge and judgements in a final paper (Communication).

Course Content

This course will cover theory, methods and techniques for:

- personalization for Web applications;
- Web user & context modeling;
- user-generated content and metadata;

- multi-device interaction;
- usage of social-web data.

The course takes a data science approach, using Jupyter notebooks to collect and analyse data, with a research paper describing the data science study as the main outcome of the course.

Additional Information Teaching Methods

Lectures (1x week)
Practical sessions (1x week)
Presentation seminars (1x week)

Method of Assessment

The weighted average of:

- 70% research paper (group assignment A),
- 10% group presentation (P),
- and 20% individual exam (E).

A minimum grade of 5.5 for each component is required.

Compensation: it is not possible to compensate one component with another.

Mode of re-examination: if insufficient, each component may be revised and resubmitted once.

Literature

- Course lecture slides;
- Selected articles, videos and Web links

All materials for each lecture will be provided through Canvas.

Additional Information Target Audience

Master Artificial Intelligence
Master Computer Science
Master Information Sciences
Master Computer Security

Additional Information

The course has a maximum enrolment limit of 120. The students of the Master Information Sciences, for whom it is compulsory, have priority over the students that take this course as an elective or a constrained choice course.

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Recommended background knowledge

Basic Python programming skills.